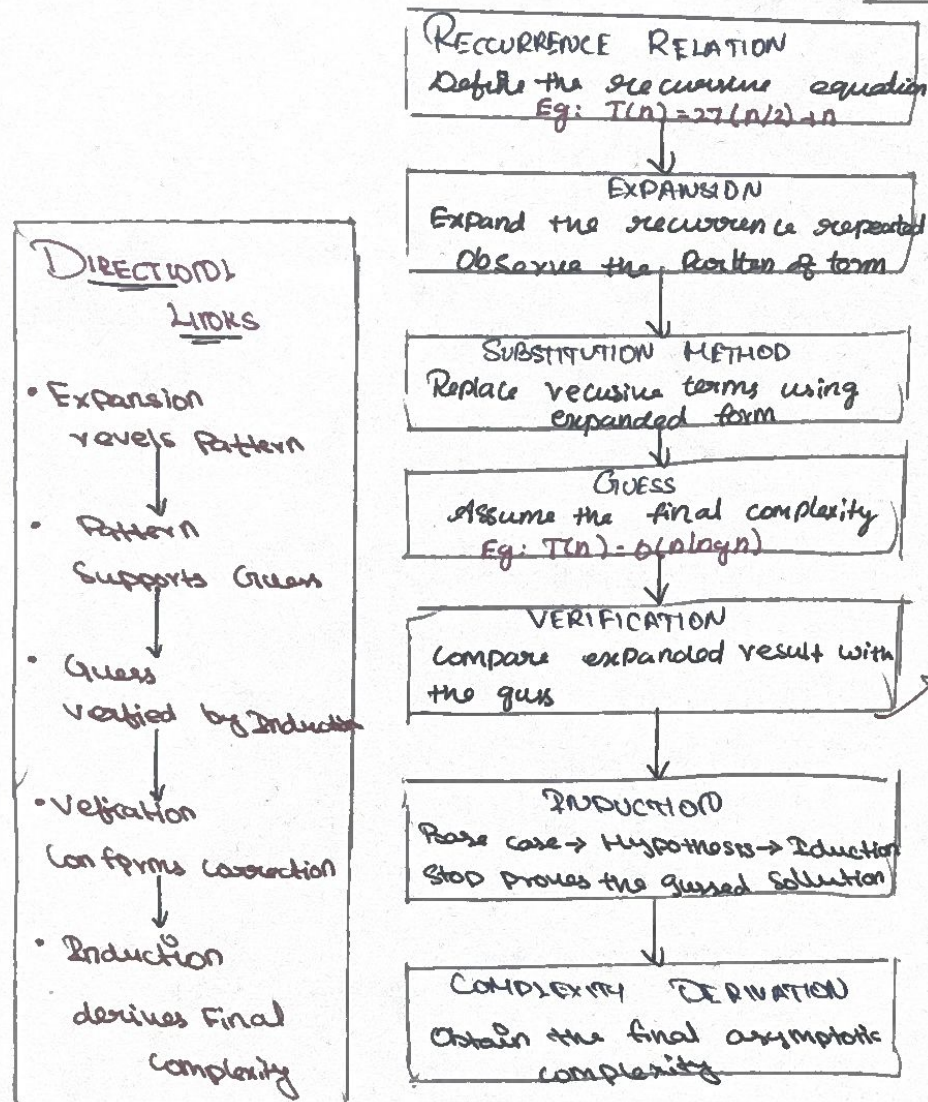


CONCEPT MAP 1: RECURRENCE RELATION $\rightarrow$ EXPANSION $\rightarrow$ SUBSTITUTION $\rightarrow$ GUESS $\rightarrow$ VERIFICATION $\rightarrow$ COMPLEXITY DERIVATION

SOLVING RECURRENCE RELATIONS USING THE SUBSTITUTION METHOD



INTERPRETATION

The concept map provides a clear step-by-step framework for solving recurrence relation. Starting from the recurrence equation, repeated substitution and expansion reveal a pattern.

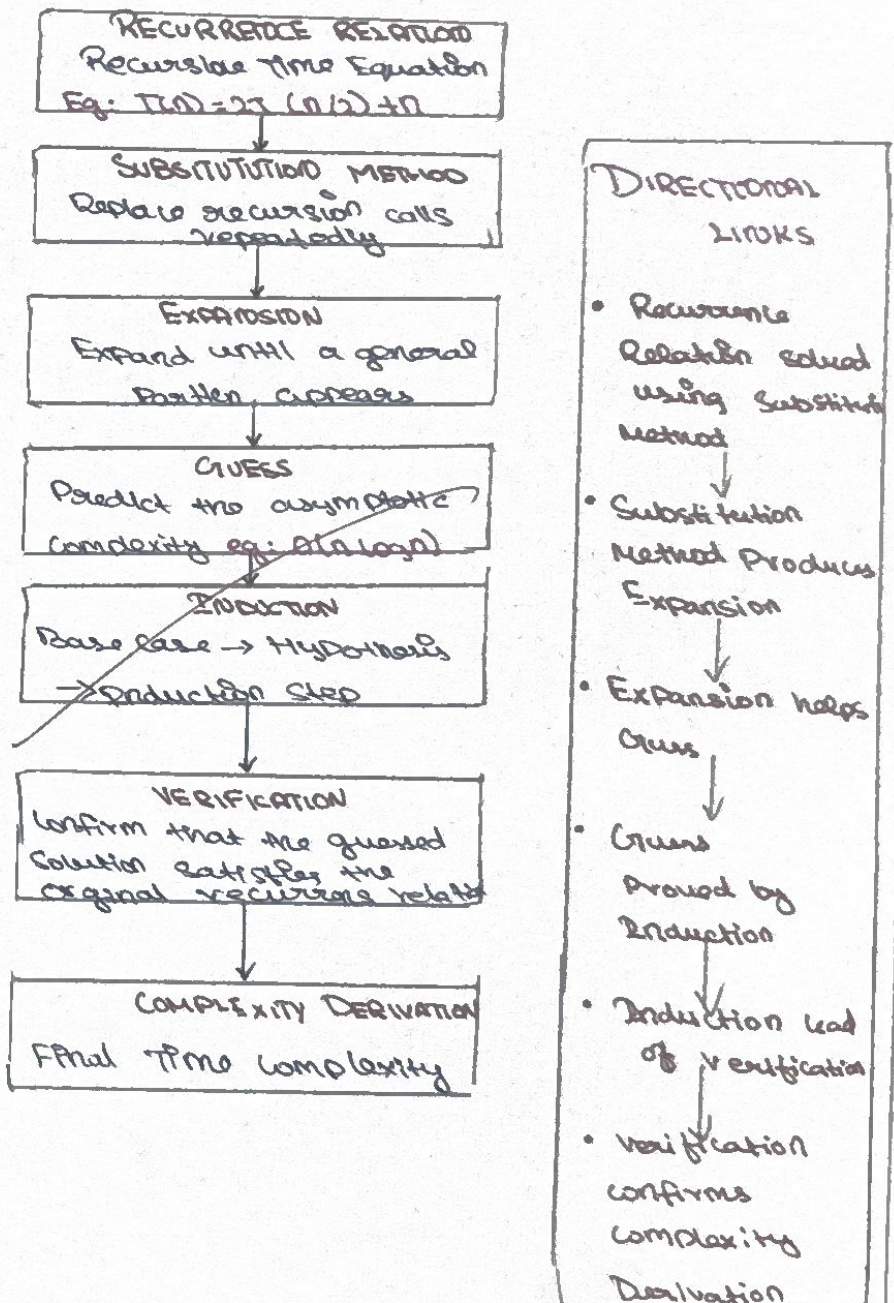
INTERPRETATION

The concept map presents a systematic procedure for solving recurrence's expanded substituted repeatedly and a complexity's guessed.

Verification and mathematical induction confirm.

CONCEPT MAP 2: LOGICAL FLOW OF THE SUBSTITUTION METHOD FOR COMPLEXITY DERIVATION

85%



KEY IDEA $\Rightarrow$ GUESS $\rightarrow$ VERIFY $\rightarrow$ PROVE BY INDUCTION $\rightarrow$ DERIVE FINAL COMPLEXITY

KEY IDEA: GUESS $\rightarrow$ PROVE BY INDUCTION $\rightarrow$ VERIFY $\rightarrow$ DERIVE FINAL COMPLEXITY